UNIVERSITY OF RUHUNA

Faculty of Engineering

EC7204: Cloud Computing

Large-Scale Data Analysis Using MapReduce

Global Terrorism Database Casualty Analysis

Group 10

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01). Approach and Methodology

This project implements a Hadoop MapReduce job to perform large-scale terrorism casualty analysis on the Global Terrorism Database (GTD) sourced from Kaggle. The GTD contains 181,691 records spanning 1970–2017 with 135 columns, and totals approximately 162 MB in raw CSV form. The MapReduce job was written in Python using Hadoop Streaming and executed on a Hadoop 3.3.6 pseudo-distributed single-node cluster running on Ubuntu/WSL. The job analyses how different attack types contribute to global casualties by computing five aggregate metrics per attack category.
Key fields used:

- attacktype1_txt - categorical attack type label
- nkill - number of fatalities per incident
- nwound - number of wounded per incident

The MapReduce job consisted of the following components:

1. Mapper (mapper.py) - Each CSV row is parsed and the mapper emits key-value pairs in the form AttackType, (kills, wounds, 1). Rows with missing or null attack-type values are skipped to ensure data quality.
2. Reducer (reducer.py) - The reducer accumulates total incidents, total kills, total wounds, and derives average kills and average wounds per incident for each attack category.

02). Results and Summary

The results reveal significant variation in lethality across attack types. Bombing and Explosion is the most frequent attack type with over 88,000 incidents and also accounts for the highest total casualties, reflecting its indiscriminate nature. Armed Assault ranks second in frequency and shows a high average kill rate of approximately 2.09 per incident, suggesting it is highly lethal when executed. Assassination attacks, while fewer in number, show a notably high kill-to-wound ratio, indicating their targeted and

precise nature. Overall, the data shows that the most frequent attack types are not always the most lethal on a per-incident basis, which has important implications for counter-terrorism resource allocation.

Below figure shows final output of the Results.

```
chathza@MSI:~/gtd_mapreduce$ cd ~/gtd_mapreduce
chathza@MSI:~/gtd_mapreduce$ python3 -c "
data = open('/home/chathza/gtd_mapreduce/results.txt').readlines()
print('=' * 105)
print(f\"{'Attack Type':<40} {'Incidents':>10} {'TotalKilled':>12} {'TotalWounded':>13} {'AvgKilled':>10} {'AvgWounded':>11}\
")
print('=' * 105)
for line in data:
    parts = line.strip().split('\t')
    if len(parts) < 6: continue
    name = parts[0]
    vals = [p.split(':')[1] for p in parts[1:]]
    print(f'{name:<40} {vals[0]:>10} {vals[1]:>12} {vals[2]:>13} {vals[3]:>10} {vals[4]:>11}')
print('=' * 105)
" > ~/gtd_mapreduce/results_clean.txt
chathza@MSI:~/gtd_mapreduce$ cat ~/gtd_mapreduce/results_clean.txt
=========================================================================================================
Attack Type                                 Incidents  TotalKilled  TotalWounded  AvgKilled  AvgWounded
=========================================================================================================
Armed Assault                                   42669       160297           482       3.76        0.01
Assassination                                   19312        24920            78       1.29        0.00
Bombing/Explosion                               88255       157321          2503       1.78        0.03
Facility/Infrastructure Attack                  10355         3642           342       0.35        0.03
Hijacking                                         659         3718            48       5.64        0.07
Hostage Taking (Barricade Incident)               991         4478           149       4.52        0.15
Hostage Taking (Kidnapping)                     11158        24231            45       2.17        0.00
Unarmed Assault                                  1015          880           876       0.87        0.86
Unknown                                          7276        32381            33       4.45        0.00
=========================================================================================================
chathza@MSI:~/gtd_mapreduce$
```

03). Performance and Accuracy Observations

The MapReduce job processed the dataset in approximately 4 - 6 minutes on a single-node cluster. The mapper phase was the bottleneck due to CSV parsing of 135 columns per row. Out of the 181,691 rows, 181,690 rows were successfully processed while one row was not processed due to a parsing issue. In a real distributed environment with multiple nodes, the mapper phase would scale linearly with the number of nodes.

04). Future Improvements

- Add a second MapReduce job to analyse trends over time (casualties per year per attack type).
- Include a geographic dimension by adding country or region as a secondary key.
- Deploy on AWS EMR or Google Dataproc for true distributed processing.
- Use Apache Spark instead of MapReduce for in-memory processing, which would be 10-100x faster.
- Build a visualization dashboard using the output data to communicate findings to policymakers.